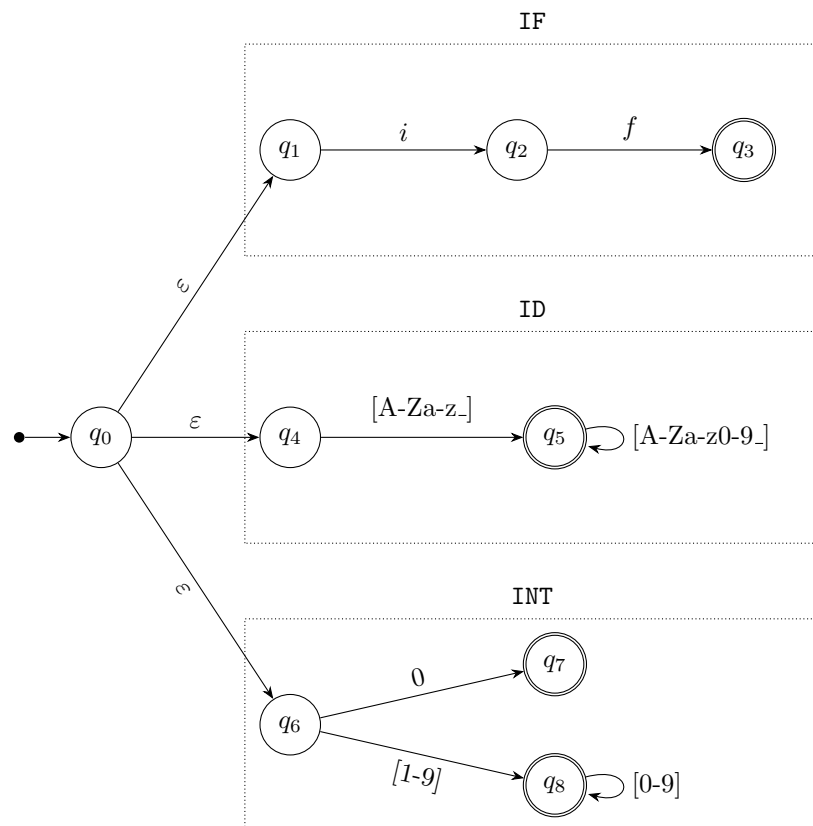


CSE411 Assignment #1: The NFA/DFA for **IF** | **ID** | **INT**

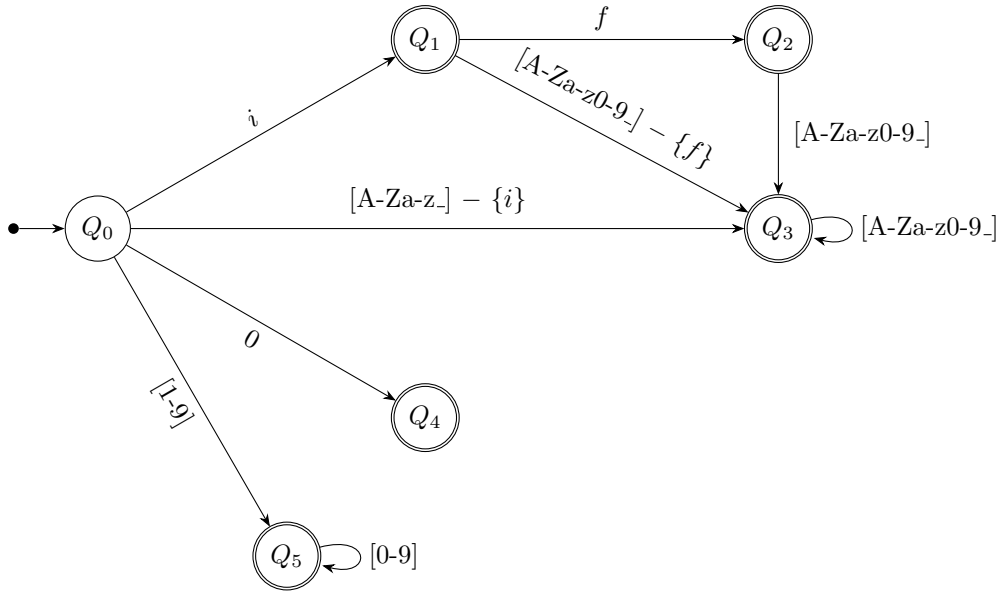
Sungho Kim (20241065)

September 14, 2026

1 NFA



2 DFA



$$Q_0 = \varepsilon\text{-closure}(\{q_0\}) = \{q_0, q_1, q_4, q_6\}.$$

DFA State D	Input Character c	DFA Transition $\delta_D(D, c)$
Q_0	i	$Q_1 = \varepsilon\text{-closure}(\{q_2, q_5\}) = \{q_2, q_5\}$
Q_1	f	$Q_2 = \varepsilon\text{-closure}(\{q_3, q_5\}) = \{q_3, q_5\}$
Q_0	$[A-Za-z_] - \{i\}$	$Q_3 = \varepsilon\text{-closure}(\{q_5\}) = \{q_5\}$
Q_1	$[A-Za-z0-9_] - \{f\}$	$Q_3 = \varepsilon\text{-closure}(\{q_5\}) = \{q_5\}$
Q_2, Q_3	$[A-Za-z0-9_]$	$Q_3 = \varepsilon\text{-closure}(\{q_5\}) = \{q_5\}$
Q_0	0	$Q_4 = \varepsilon\text{-closure}(\{q_7\}) = \{q_7\}$
Q_0	$[1-9]$	$Q_5 = \varepsilon\text{-closure}(\{q_8\}) = \{q_8\}$
Q_5	$[0-9]$	$Q_5 = \varepsilon\text{-closure}(\{q_8\}) = \{q_8\}$